

Geometric Singular Perturbation Theory in the Stable Boundary Layer: Coordinate Curvature, Turbulence Collapse, and Discrete Numerical Architecture

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Abstract

Vertical profiles of wind and temperature in the stable boundary layer (SBL) frequently exhibit sharp knees and abrupt regime transitions that challenge standard Monin–Obukhov similarity closures and numerical weather prediction (NWP) solvers. In this work, we present a unified mathematical framework grounded in Geometric Singular Perturbation Theory (GSPT) that decouples physical turbulence transitions from spatial coordinate mapping artifacts. Using Nieuwstadt local scaling re-parametrized by the inverse Obukhov length profile $\chi(z) \equiv 1/L(z)$, we prove that physical-space profile curvature ($Ri_{g,zz}$) decomposes into intrinsic closure curvature ($R''\zeta_z^2$) and coordinate-induced mapping curvature ($R'\zeta_{zz}$). Nondegenerate coordinate folds ($\zeta_z = 0$) occur strictly outside constant-flux layers, driven by differential flux divergence between momentum and heat, and generate non-zero physical profile curvature independently of closure failure. Dynamically, we model SBL turbulence collapse as a fast-slow dynamical system where turbulent kinetic energy relaxes to a fast-slow critical manifold. Transition to the decoupled, very stable regime corresponds to a saddle-node fold where the fast eigenvalue vanishes ($\lambda_f \rightarrow 0$), destroying normal hyperbolicity. To prevent numerical stiffness and spurious superlinear modal blowups in low-order discretizations, we introduce Fenichel invariant manifold slaving alongside an adaptive equidistribution grid-stretching formulation ($\Delta z \propto 1/|\zeta_{zz}|$). We prove that bounded grid stretching achieves quadratic spatial convergence $\mathcal{O}(N_z^{-2})$ while Fenichel slaving regular-

izes solver stiffness, guaranteeing global time-step stability under non-vanishing Courant–Friedrichs–Lewy (CFL) limits. Finally, we establish an asymptotic boundary regularity threshold ($p = b - \frac{3}{2}a \geq 2$) for sub-grid turbulence schemes to eliminate coordinate singularities at the SBL top.

1 Introduction

Modeling and parameterizing the stably stratified atmospheric boundary layer (SBL) remains one of the most persistent challenges in geophysical fluid dynamics, numerical weather prediction (NWP), and climate modeling [?, ?]. Under strongly stable conditions, turbulent mixing is suppressed by buoyancy forces, leading to a decoupling of the surface from the overlying atmosphere, the formation of nocturnal Low-Level Jets (LLJs), and intense surface radiative cooling [?, ?]. Standard Monin–Obukhov Similarity Theory (MOST) closures—originally derived under constant-flux surface layer assumptions—regularly break down in these highly stratified, non-stationary regimes, causing single-column models (SCMs) and Earth System Models (ESMs) to exhibit severe temperature and wind velocity biases [?, ?, ?].

To resolve these biases, developers conventionally calibrate empirical stability parameters (such as the Businger-Dyer coefficients β_m, β_h) when simulated profiles diverge from observations [?, ?, ?]. However, vertical profiles of the gradient Richardson number (Ri_g) reconstructed from meteorological towers and remote sensing campaigns frequently exhibit sharp inflection “knees” and apparent coordinate folds [?, ?].

In this work, we present a unified mathematical framework grounded in Geometric Singular Perturbation Theory (GSPT) to demonstrate that these inflections are often coordinate-mapping features ("Fold Illusions") generated by the non-linear projection of similarity coordinates under active vertical flux divergence, rather than genuine dynamical transitions of the boundary layer [?, ?].

Dynamically, SBL turbulence collapse can be modeled as a multiscale fast-slow dynamical system, where the microscale Turbulent Kinetic Energy (TKE) relaxing rapidly onto a critical manifold $\mathcal{C}_0$ governed by the slower evolution of mean vertical wind shear S [?, ?]. The scale separation between these turbulent and geostrophic subsystems is parameterized by a small dimensionless ratio $0 < \epsilon \ll 1$ [?]:

$$\epsilon \frac{dE}{dt} = f(E, S) = l\sqrt{E + \delta} \left(S^2 - \phi N^2 - c_b N^2 \frac{E}{E + \alpha} \right) \quad (1)$$

$$\frac{dS}{dt} = g(E, S) = G - c_1 ES - c_2 S, \quad (2)$$

where E represents the grid-resolved TKE, l is the turbulent mixing length, N^2 is the Brunt-Väisälä buoyancy frequency, and G represents geostrophic wind forcing [?, ?].

A long-standing computational issue in these multiscale models is the existence of a mathematical singularity in the production Jacobian $(\partial f / \partial E)$ as TKE approaches zero [?]. Standard closures closed with a velocity scale of $\sqrt{E}$ suffer from infinite derivative growth ($\lim_{E \rightarrow 0^+} \frac{\partial}{\partial E}(l\sqrt{E}S^2) = +\infty$), triggering severe numerical instabilities near the laminar state [?]. To restore mathematical regularity and ensure a globally C^1 -continuous fast vector field on the physically admissible domain $\mathcal{D} = \{(E, S) \in \mathbb{R}^2 : E \geq 0\}$, we embed a velocity-scale regularization floor $\delta = 10^{-6} \text{ m}^2\text{s}^{-2}$ directly into the shear production and buoyant loss terms of the TKE budget [?, ?].

By formulating SBL transitions as slow-passage bifurcations across a regularized critical manifold, we establish a mathematically consistent, singularity-free boundary-layer architecture. This paper is organized as follows: Section 2 derives the continuous differential geometry of SBL coordinate curvature under Nieuwstadt local scaling; Section 3 analyzes the topological bifurcations and hysteresis of the non-monotonic buoyancy closure; Section 4 provides mathematically rigorous proofs for discrete grid spacing and Fenichel manifold slaving convergence; Section 5 outlines the operational implications for SCM and

NWP error partitioning; and Section 6 synthesizes our findings and details future research avenues.

2 GSPT Coordinate Curvature and Nieuwstadt Local Scaling

Evaluating vertical profiles of atmospheric stability beyond the shallow surface layer requires abandoning the constant-flux assumption of classical Monin-Obukhov Similarity Theory [?]. In the outer stable boundary layer, vertical turbulent transport decays continuously, causing the turbulent heat and momentum fluxes to vary with height [?, ?]. To account for this vertical structure, we invoke Nieuwstadt's local scaling hypothesis [?]. Under this local scaling framework, the Monin-Obukhov length is generalized to a dynamically varying, height-dependent physical scaling length $L(z)$, defined as [?]:

$$L(z) \equiv \frac{-\tau(z)^{3/2}}{\kappa \frac{g}{\theta_0} H(z)}, \quad (3)$$

where $\tau(z) > 0$ represents the local vertical shear stress (momentum flux), $H(z) \equiv \overline{w'\theta_v'}(z) < 0$ represents the local kinematic virtual potential temperature flux, $\kappa \approx 0.40$ is the von Kármán constant, and g/θ_0 is the buoyancy parameter [?].

To map this height-dependent scaling without introducing numerical singularities near neutral conditions where fluxes approach zero and $L(z) \rightarrow \pm\infty$, we define the *inverse Obukhov length profile*, $\chi(z) \equiv 1/L(z)$ [?]. The local dimensionless similarity coordinate $\zeta(z)$ is mapped as:

$$\zeta(z) \equiv \frac{z}{L(z)} = z\chi(z). \quad (4)$$

Assuming active turbulence ($\tau > 0$) and that the inverse profile is twice-differentiable ($\chi \in C^2$), we compute the first two vertical spatial derivatives of the similarity coordinate mapping with respect to physical height z :

$$\zeta_z(z) = \chi(z) + z\chi'(z), \quad (5)$$

$$\zeta_{zz}(z) = 2\chi'(z) + z\chi''(z). \quad (6)$$

By expanding the logarithmic derivative of the inverse Obukhov profile, we express the coordinate Jacobian $\zeta_z(z)$ directly in terms of the relative vertical flux gradients of heat and momentum:

$$\zeta_z(z) = \chi(z) \left[1 + z \left(\frac{H'(z)}{H(z)} - \frac{3}{2} \frac{\tau'(z)}{\tau(z)} \right) \right]. \quad (7)$$

2.1 Kinematic Folds and Structural Exclusion

A *coordinate fold point* $z_* \in (0, z_{\text{top}})$ represents a physical vertical height where the coordinate Jacobian vanishes ($\zeta_z(z_*) = 0$) [?]. The fold is non-degenerate if the coordinate curvature is non-zero ($\zeta_{zz}(z_*) \neq 0$). Under Nieuwstadt local scaling, coordinate folds are physically prohibited within constant-flux layers:

Proposition 1 (Structural Exclusion of Folds). *Let $J \subset (0, z_{\text{top}})$ be a physical vertical subinterval on which the turbulent fluxes are spatially uniform, such that the inverse Obukhov profile is constant ($\chi(z) \equiv \chi_0 \neq 0$, representing a classical constant-flux surface layer). Then, the coordinate Jacobian strictly satisfies $\zeta_z(z) = \chi_0 \neq 0$ for all $z \in J$; hence, J contains no coordinate fold points.*

Proof. On the subinterval J , the vertical flux gradients vanish identically, $\chi'(z) \equiv 0$ [?]. Substituting this into the coordinate Jacobian equation (??) yields $\zeta_z(z) = \chi(z) + z(0) = \chi_0$ [?]. Since $\chi_0 \neq 0$, the Jacobian never vanishes on J , preventing the existence of coordinate folds [?]. $\square$

Proposition 1 proves that the existence of an observed coordinate fold is a kinematic signature of active vertical flux divergence. Setting $\zeta_z(z_*) = 0$ in (??) yields the generic coordinate fold condition:

$$1 + z_* \left(\frac{H'(z_*)}{H(z_*)} - \frac{3}{2} \frac{\tau'(z_*)}{\tau(z_*)} \right) = 0. \quad (8)$$

Equation (??) proves that a coordinate fold in a stable, turbulent layer is governed by the *differential flux divergence* between momentum and heat, rather than thermal flux divergence alone.

2.2 The GSPT Curvature Decomposition

Let $Ri_g(z) = R(\zeta(z))$ represent the gradient Richardson number, where $R(\cdot)$ is a smooth similarity closure in similarity space. Differentiating the Richardson profile twice with respect to physical height z yields the GSPT curvature decomposition:

$$Ri_{g,z} = R'(\zeta)\zeta_z, \quad (9)$$

$$Ri_{g,zz} = \underbrace{R''(\zeta)\zeta_z^2}_{C_{\text{constitutive}}} + \underbrace{R'(\zeta)\zeta_{zz}}_{C_{\text{mapping}}}. \quad (10)$$

Equation (??) partitions the total profile curvature into two additive, mathematically distinct geometric objects:

1. **Constitutive Curvature** ($C_{\text{constitutive}}$): Represents the physical curvature of the empirical stability relation mapped into physical space. Because it scales quadratically with the coordinate Jacobian (ζ_z^2), it remains highly sensitive to local coordinate stretching.
2. **Mapping Curvature** (C_{mapping}): Represents the pure coordinate-induced curvature generated by the non-linear projection $z \mapsto \zeta(z)$ under active vertical flux divergence ($\zeta_{zz} \neq 0$).

Theorem 1 (Curvature Signature of a Nondegenerate Fold). *Let z_* represent a nondegenerate coordinate fold point ($\zeta_z(z_*) = 0$ and $\zeta_{zz}(z_*) \neq 0$) under stable stratification, and assume $R'(\zeta_*) \neq 0$. Then, the physical gradient Richardson profile exhibits a stationary point with non-zero vertical curvature governed entirely by the mapping coordinate geometry:*

$$Ri_{g,z}(z_*) = 0 \quad \text{and} \quad Ri_{g,zz}(z_*) = R'(\zeta_*)\zeta_{zz}(z_*) \neq 0. \quad (11)$$

Proof. Evaluating the first derivative (??) at z_* where $\zeta_z(z_*) = 0$ yields $Ri_{g,z}(z_*) = R'(\zeta_*)(0) = 0$ [?]. Evaluating the GSPT curvature decomposition (??) at the fold yields $Ri_{g,zz}(z_*) = R''(\zeta_*)(0)^2 + R'(\zeta_*)\zeta_{zz}(z_*) = R'(\zeta_*)\zeta_{zz}(z_*)$ [?]. Since $R'(\zeta_*) \neq 0$ and nondegeneracy guarantees $\zeta_{zz}(z_*) \neq 0$, the total physical vertical curvature at the fold remains strictly non-zero [?]. $\square$

To quantify whether an observed vertical spatial "knee" ($|Ri_{g,zz}| \gg 0$) stems from similarity closure failure or coordinate mapping curvature, we define the *coordinate-induced curvature ratio*, $\mathcal{C}_{\text{coord}}(z)$:

$$\mathcal{C}_{\text{coord}}(z) \equiv \frac{|R'(\zeta)\zeta_{zz}|}{|R''(\zeta)\zeta_z^2| + |R'(\zeta)\zeta_{zz}| + \varepsilon_C}, \quad (12)$$

where $\varepsilon_C > 0$ is a small regularization scale. A value of $\mathcal{C}_{\text{coord}} \rightarrow 1$ certifies that the apparent physical profile knee is a "Fold Illusion" driven by non-linear coordinate mapping, whereas $\mathcal{C}_{\text{coord}} \rightarrow 0$ indicates genuine structural curvature in the stability relation $R(\zeta)$.

3 Fast-Slow SBL Manifold Bifurcations

Standard boundary-layer closures assume monotonic buoyant damping ($d\mathcal{B}/dE > 0$), forcing a single continuous equilibrium that prevents models from capturing the sudden, discontinuous turbulence collapse

observed during nocturnal transitions [?]. To model this regime transition dynamically, we introduce a non-monotonic, hump-shaped buoyancy-destruction closure, $D_b(E; N^2)$, representing the non-linear saturation of buoyancy fluxes under strongly stable stratification [?]:

$$D_b(E; N^2) = c_b N^2 \frac{\alpha E}{(E + \alpha)^2},$$

where $c_b > 0$ is a dimensionless efficiency factor and $\alpha > 0$ represents the regularized energy scale [?]. Differentiating (??) with respect to TKE yields:

$$D'_b(E; N^2) = c_b N^2 \alpha \frac{\alpha - E}{(E + \alpha)^3}. \quad (14)$$

Equation (??) reveals two physically distinct regimes:

- **Sub-critical mixing** ($E \leq \alpha$): As TKE decreases, negative buoyancy fluxes efficiently quench vertical velocity variance ($D'_b(E) > 0$), dampening turbulence [?].
- **Super-critical mixing** ($E > \alpha$): At higher TKE, energetic turbulent eddies break up localized stratification structures, suppressing the relative efficiency of buoyant destruction ($D'_b(E) < 0$) [?].

Under the corrected 1.5-order prognostic TKE budget, the fast subsystem satisfies [?]:

$$\epsilon \frac{dE}{dt} = f(E, S; N^2) = l\sqrt{E + \delta} \left[S^2 - \phi N^2 - c_b N^2 \frac{\alpha E}{(E + \alpha)^2} \right] \quad (15)$$

where $\delta = 10^{-6} \text{ m}^2\text{s}^{-2}$ is the singularity regularization floor [?]. Setting the TKE tendency to zero ($f(E, S) = 0$) defines the critical manifold $\mathcal{C}_0$ as the equilibrium wind shear surface, $S(E; N^2)$ [?]:

$$S(E; N^2) = \sqrt{\phi N^2 + c_b N^2 \frac{\alpha E}{(E + \alpha)^2} + \frac{E^{3/2}}{l^2 \sqrt{E + \delta}}}. \quad (16)$$

An ordinary saddle-node fold point (E^*, S^*, N^{*2}) occurs where the equilibrium curve has a turning point ($dS/dE = 0$), which corresponds to the simultaneous equations [?]:

$$f(E, S; N^2) = 0 \quad \text{and} \quad f_E(E, S; N^2) \equiv \lambda_f = 0, \quad (17)$$

where the fast-manifold stability eigenvalue λ_f is defined as [?]:

$$\lambda_f = \frac{l(S^2 - \phi N^2)}{2\sqrt{E + \delta}} - c_b N^2 l \left[\frac{E}{2\sqrt{E + \delta}(E + \alpha)} + \frac{\alpha\sqrt{E + \delta}}{(E + \alpha)^2} \right] - \frac{3\sqrt{E}}{2l} \quad (18)$$

3.1 Analytical Derivation of the Fold Locus

Substituting the equilibrium condition (??) directly into the zero-eigenvalue condition (??) eliminates the mean shear variable S , mapping the exact regularized *Fold Locus* as a function of TKE [?, ?]:

$$N_{\text{fold}}^2(E; \delta) = \frac{(E + \alpha)^3 E^{1/2} (3E + 4\delta)}{2l^2 c_b \alpha (E - \alpha) (E + \delta)^{3/2}} \quad \forall E > \alpha. \quad (19)$$

Equation (??) converts the two-dimensional fold search into a one-dimensional geometric minimization problem. Because the denominator contains $(E - \alpha)$, a physical fold (where $N_{\text{fold}}^2 > 0$) strictly requires $E > \alpha$, proving that folds cannot occur within the sub-critical mixing regime [?].

3.2 The Critical Cusp and Stratification Threshold

The fold pair appears when the fold locus reaches its mathematical minimum with respect to E . We define this threshold as the *critical stratification*, $N_{\text{crit}}^2 \equiv \min_{E > \alpha} N_{\text{fold}}^2(E; \delta)$ [?]. In the singular unregularized analytical limit where $\delta \rightarrow 0$, taking the logarithmic derivative of the fold locus and setting it to zero

$$\frac{d}{dE} \left[\frac{3(E + \alpha)^3}{2l^2 c_b \alpha (E - \alpha)} \right] = 0 \quad \implies \quad \frac{3(E + \alpha)^2 (E - 2\alpha)}{(E - \alpha)^2} = 0. \quad (20)$$

The unique physical root of (??) establishes the exact cusp coordinate, $E_{\text{cusp}} = 2\alpha$ [?]. Substituting this cusp coordinate back into the unregularized fold locus yields the exact critical cusp stratification [?]:

$$N_{\text{crit}}^2(0) = \frac{81\alpha}{2l^2 c_b}. \quad (21)$$

For our default parameters ($l = 1.0 \text{ m}$, $c_b = 0.50$, $\alpha = 0.05 \text{ m}^2\text{s}^{-2}$), this analytical limit resolves to $E_{\text{cusp}} = 0.10 \text{ m}^2\text{s}^{-2}$ and $N_{\text{crit}}^2 \approx 4.05 \text{ s}^{-2}$ [?]. Incorporating finite regularization ($\delta = 10^{-6} \text{ m}^2\text{s}^{-2}$) slightly perturbs this minimum but preserves the underlying topology [?].

3.3 Topological Manifold Branches and Regime Transitions

For stratifications exceeding the critical threshold ($N^2 > N_{\text{crit}}^2$), the equilibrium manifold contains a folded, three-branch structure [?, ?]:

1. **The Weakly Stable Boundary Layer (WSBL) branch** ($E > E_{\text{fold,upper}}$): An attracting branch ($\lambda_f < 0$) characterized by high TKE, strong shear production, and continuous vertical mixing [?].
2. **The Fast-Repelling branch** ($E_{\text{fold,lower}} < E < E_{\text{fold,upper}}$): An unstable branch ($\lambda_f > 0$) that acts as the state-space separatrix/basin boundary between regimes [?].
3. **The Very Stable Boundary Layer (VSBL) branch** ($E < E_{\text{fold,lower}}$): An attracting branch ($\lambda_f < 0$) characterized by suppressed turbulence, weak surface coupling, and potential for runaway nocturnal cooling [?].

The two folds represent static transition thresholds where the local stable equilibrium terminates, forcing the boundary layer trajectory to jump abruptly to the other attracting branch [?].

4 Discrete Numerical Architecture and Stability Proofs

To translate the continuous geometric and dynamical formulations of GSPT into discrete atmospheric models without introducing unphysical grid-point coalescence or explicit time-step crashes, we establish an adaptive grid-stretching algorithm alongside an invariant manifold solver [?].

4.1 Discrete Equidistribution and Bounded Grid Stretching

Let $z \in [0, H_{\text{SBL}}]$ represent physical vertical height, and let $\zeta_{zz}(z)$ denote the vertical coordinate curvature under local Nieuwstadt scaling. We define the adaptive physical coordinate density function $\rho_{\text{clamp}}(z)$ as [?]:

$$\rho_{\text{clamp}}(z) \equiv \max \left(\frac{k}{\Delta z_{\text{max}}}, \min \left(\frac{k}{\Delta z_{\text{min}}}, |\zeta_{zz}(z)| + \epsilon_z \right) \right) \quad (22)$$

where Δz_{min} and Δz_{max} represent the minimum and maximum vertical grid steps, $\epsilon_z > 0$ is a coordinate

regularization parameter, and $k > 0$ is a normalization scalar. Let $C(z) = \int_0^z \rho_{\text{clamp}}(s) ds$ define the cumulative density function. The physical vertical levels z_j (for $j = 1, \dots, N_z$) are generated by uniform equidistribution of the computational coordinate $y \in [0, 1]$ [?]:

$$y_j = \frac{j-1}{N_z-1}, \quad z_j = y^{-1}(y_j). \quad (23)$$

Theorem 2 (Discrete Grid-Spacing Bounds). *Under the bounded equidistribution density (??), the discrete physical grid steps $\Delta z_j \equiv z_j - z_{j-1}$ are globally bounded by:*

$$\Delta z_{\text{min}} \leq \Delta z_j \leq \Delta z_{\text{max}} \quad \forall j \in \{2, \dots, N_z\}. \quad (24)$$

Proof. Applying the Mean Value Theorem to the normalized cumulative density function $y(z)$ on the interval $[z_{j-1}, z_j]$ guarantees an intermediate point $\xi \in (z_{j-1}, z_j)$ such that [?]:

$$y(z_j) - y(z_{j-1}) = y'(\xi)(z_j - z_{j-1}) \implies \frac{1}{N_z - 1} = \frac{\rho_{\text{clamp}}(\xi)}{C_{\text{max}}} \Delta z_j \quad (25)$$

Solving for the local grid step yields $\Delta z_j = \frac{C_{\text{max}}}{(N_z - 1)\rho_{\text{clamp}}(\xi)}$ [?]. By definition, the clamped density is bounded pointwise by $\frac{k}{\Delta z_{\text{max}}} \leq \rho_{\text{clamp}}(z) \leq \frac{k}{\Delta z_{\text{min}}}$ [?]. Integrating this across the compact domain yields the bounds on the cumulative mass:

$$\frac{kH_{\text{SBL}}}{\Delta z_{\text{max}}} \leq C_{\text{max}} \leq \frac{kH_{\text{SBL}}}{\Delta z_{\text{min}}}. \quad (26)$$

Setting the computational scaling factor to $k \equiv \frac{C_{\text{max}}}{N_z - 1}$, the grid-spacing relation simplifies to $\Delta z_j = \frac{k}{\rho_{\text{clamp}}(\xi)}$ [?]. Substituting the pointwise density bounds into this expression yields:

$$\Delta z_{\text{min}} \leq \Delta z_j \leq \Delta z_{\text{max}}, \quad (27)$$

completing the proof. $\square$

Theorem 3 (Asymptotic Grid Convergence). *Let $\chi(z) \in C^2$ represent a smooth, twice-differentiable inverse Obukhov length profile. Then, the discrete vertical coordinates z_j converge asymptotically to the continuous optimal coordinate path $z(y)$ with quadratic accuracy in the vertical level resolution:*

$$\max_{1 \leq j \leq N_z} |z_j - z(y_j)| = \mathcal{O}(N_z^{-2}). \quad (28)$$

Proof. Let $z(y)$ define the continuous inverse coordinate function, satisfying $y(z(y)) = y$. Since $\chi(z) \in C^2$, the coordinate density $\rho_{\text{clamp}}(z)$ is C^1 -continuous on the compact domain $[0, H_{\text{SBL}}]$ [?]. Applying a second-order Taylor expansion to the inverse mapping $z(y)$ about the computational node y_j yields [?]:

$$z(y_j) = z(y_{j-1}) + \frac{dz}{dy}(y_{j-1})\Delta y + \frac{1}{2} \frac{d^2z}{dy^2}(\eta_j)(\Delta y)^2, \quad (29)$$

where $\Delta y = \frac{1}{N_z-1}$ and $\eta_j \in (y_{j-1}, y_j)$ [?]. Using the inverse derivative relations, we calculate:

$$\frac{dz}{dy} = \frac{C_{\text{max}}}{\rho_{\text{clamp}}(z)}, \quad \frac{d^2z}{dy^2} = -\frac{C_{\text{max}}^2 \rho'_{\text{clamp}}(z)}{[\rho_{\text{clamp}}(z)]^3} \frac{dz}{dy}. \quad (30)$$

Because the clamped density is bounded away from zero ($\rho_{\text{clamp}} \geq \epsilon_z > 0$) and its derivatives are bounded, the second derivative is bounded pointwise by a positive constant $M > 0$ [?]. The global discretization error is bounded by:

$$\max_{1 \leq j \leq N_z} |z_j - z(y_j)| \leq \frac{1}{2} \max_{y \in [0, 1]} \left| \frac{d^2z}{dy^2} \right| (\Delta y)^2 \leq \frac{M}{2(N_z - 1)^2} \quad (31)$$

completing the proof. $\square$

4.2 Fenichel Invariant Manifold Slaving and Stiffness Regularization

Low-order Galerkin modal projections of non-linear boundary-layer budgets are prone to spurious, superlinear finite-time blowup ($t_{\text{blowup}} \leq \frac{1}{2\sigma u_1(0)^2}$) near turning points, where the fast TKE eigenvalue vanishes ($\lambda_f \rightarrow 0$) [?]. This instability occurs because modal truncation allows the secondary fast mode u_2 to undergo negative excursions ($u_2 \propto -u_1^2$), introducing a positive cubic feedback term (σu_1^3) into the primary slow mode equation [?].

To regularize this stiffness, we apply *Fenichel Invariant Manifold Slaving* [?]. We replace the prognostic ODE for the secondary mode, $\dot{u}_2$, with the regularized algebraic slow-manifold parameterization [?]:

$$u_2 = \Phi(u_1, S) \equiv -\frac{c_{\text{cross}} u_1^2}{2|\lambda_f(S)| + \delta_m}, \quad (32)$$

where $c_{\text{cross}} > 0$ is the spectral cross-coupling coefficient and $\delta_m \sim 10^{-3} \text{ s}^{-1}$ is a hyperbolic regularization parameter that bounds the manifold solver near the fold locus where the fast eigenvalue vanishes [?].

Theorem 4 (Global Boundedness of Slaved Galerkin Trajectories). *The algebraic substitution of the regularized Fenichel slaving relation (??) into the primary slow mode equation eliminates the spurious cubic production feedback term σu_1^3 entirely, establishing a globally bounded, stable attractor on the slow manifold $\mathcal{S}_\epsilon$.*

Proof. Let the primary slow shear mode u_1 satisfy the governing modal ODE [?]:

$$\dot{u}_1 = G - c_{\text{shear}} u_1 - \beta u_1 u_2 - K_d u_1^3, \quad (33)$$

where $G > 0$ is geostrophic forcing, $c_{\text{shear}} > 0$ is linear damping, $\beta > 0$ is coupling, and $K_d > 0$ is physical spectral dissipation [?]. Substituting the Fenichel slaving relation (??) into (??) yields the closed system:

$$\dot{u}_1 = G - c_{\text{shear}} u_1 + \left(\frac{\beta c_{\text{cross}}}{2|\lambda_f(S)| + \delta_m} - K_d \right) u_1^3. \quad (34)$$

Let $\mu(S) \equiv \frac{\beta c_{\text{cross}}}{2|\lambda_f(S)| + \delta_m}$ represent the regularized production feedback rate. Because the positive regularization threshold $\delta_m > 0$ bounds the denominator from below, we have $\mu(S) \leq \frac{\beta c_{\text{cross}}}{\delta_m} < \infty$ globally [?]. By choosing a physical spectral dissipation parameter K_d that satisfies the sub-critical dissipation constraint:

$$K_d > \frac{\beta c_{\text{cross}}}{\delta_m}, \quad (35)$$

the effective cubic coefficient remains strictly negative for all shear states: $\Lambda(S) \equiv \mu(S) - K_d \leq -\Lambda_0 < 0$ [?]. The dynamical system is bounded by the differential inequality $\dot{u}_1 \leq G - c_{\text{shear}} u_1 - \Lambda_0 u_1^3$, defining a compact, invariant trapping region:

$$\mathcal{K} = \left\{ u_1 \in \mathbb{R} : 0 \leq u_1 \leq \sqrt{\frac{G}{\Lambda_0}} + 1 \right\}. \quad (36)$$

Any trajectory starting outside $\mathcal{K}$ has a strictly negative derivative ($\dot{u}_1 < 0$) and enters $\mathcal{K}$ in finite time, where it remains trapped for all future times ($t > 0$) [?]. This completes the proof. $\square$

4.3 Coupled Solver CFL Guarantees

In production NWP applications, the maximum allowable integration step Δt is governed by the Courant-Friedrichs-Lewy (CFL) limit of the vertical diffusion solver [?]. Under our bounded grid-stretching and Fenichel slaving solver, we guarantee that the coupled solver remains stable under a non-vanishing minimum time step:

Proposition 2 (Global CFL Stability Guarantee). *The maximum time step Δt required to maintain numerical stability in the coupled TKE-shear solver remains strictly bounded from below by a non-zero physical limit, independent of the singular perturbation limit $\epsilon \rightarrow 0$ or the zero-shear limit $S \rightarrow 0$:*

$$\Delta t \geq \frac{(\Delta z_{\min})^2}{2K_{\max}} \equiv \Delta t_{\text{CFL}} > 0, \quad (37)$$

where $K_{\max}$ is the maximum turbulent exchange coefficient bounded by the slow-manifold TKE trapping region.

Proof. An explicit vertical diffusion scheme requires $\Delta t \leq \frac{(\Delta z_j)^2}{2K_j}$ pointwise [?]. By Theorem 3, the discrete physical grid spacing is bounded from below by the clamped grid generator: $\Delta z_j \geq \Delta z_{\min} > 0$ [?]. Furthermore, because Theorem 5 establishes a globally bounded TKE ($E \leq E_{\max}$), the vertical turbulent exchange coefficients (which scale as $K = l\sqrt{E + \delta}$) are globally bounded from above by $K \leq K_{\max} < \infty$ [?]. This keeps the local explicit integration limit bounded away from zero: $\Delta t \geq \frac{(\Delta z_{\min})^2}{2K_{\max}} > 0$, completing the proof. $\square$

5 SCM Evaluation and NWP Algorithmic Implications

Implementing these continuous geometric and dynamical formulations within operational SCM and NWP architectures provides three critical algorithmic fixes to prevent numerical artifacts and solver instabilities near stable boundary layer transitions.

5.1 Operational Curvature Residual Partitioning

To prevent SCM developers from committing compensating errors by "tuning" the empirical Monin-Obukhov constants (β_m, β_h) to resolve physical-space profile misfits, we establish an additive, testable curvature error partition on the tower grid [?, ?]:

$$\Delta_{\text{SCM}}(z) \equiv R_{zz}^{\text{model}}(z) - R_{zz}^{\text{obs}}(z) = \Delta_{\text{closure}}(z) + \Delta_{\text{geom}}(z) + \tau_{\Delta z}(z), \quad (38)$$

where each isolated residual component is defined operationally relative to a reference coordinate mapping [?]:

1. **Constitutive Closure Failure (Δ_{closure}):** Projects the model's similarity relation,

$R^{\text{model}}(\zeta)$, onto the observed, well-conditioned coordinate mapping, isolating local stability parameterization errors under identical physical forcing:

$$\Delta_{\text{closure}}(z) \equiv \left(R_{\zeta\zeta}^{\text{model}}(\zeta^{\text{obs}}) [\zeta_z^{\text{obs}}]^2 + R_{\zeta}^{\text{model}}(\zeta^{\text{obs}}) \zeta_{zz}^{\text{obs}} \right). \quad (39)$$

2. Flux-Coordinate Geometric Distortion (Δ_{geom}):

Projects the observed similarity closure, $R^{\text{obs}}(\zeta)$, onto the model's simulated coordinate mapping, isolating vertical transport and flux-divergence curvature errors:

$$\Delta_{\text{geom}}(z) \equiv \left(R_{\zeta\zeta}^{\text{obs}}(\zeta^{\text{model}}) [\zeta_z^{\text{model}}]^2 + R_{\zeta}^{\text{obs}}(\zeta^{\text{model}}) \zeta_{zz}^{\text{model}} \right) \quad (40)$$

3. Discretization and Grid-Scale Truncation Artifact ($\tau_{\Delta z}$):

Reconstructs the vertical truncation error introduced by the model's coarse vertical grid layout using the fourth spatial derivative of our regularized primitive splines:

$$\tau_{\Delta z}(z) \approx \frac{(\Delta z)^2}{12} \frac{\partial^4 R_{ig}}{\partial z^4}. \quad (41)$$

SCM developers can utilize these partitioned residuals to systematically isolate the true source of profile errors: if $|\Delta_{\text{geom}}| \gg 0$, developers must retune vertical mixing lengths (l_m) or transport closures while leaving the empirical similarity functions (ϕ_m, ϕ_h) completely unaltered [?].

5.2 SCM Calibration Gradient Masking

To prevent parameter drift during automated SCM optimization (e.g., in variational data assimilation or machine learning parameter estimation), the diagnostic ratio $\mathcal{C}_{\text{coord}}(z)$ must be converted into a dynamic weighting function $w(z)$ for the loss gradient with respect to the similarity parameters θ in the stability function $R(\zeta; \theta)$ [?]:

$$w(z) = 1 - \tanh(\gamma \cdot \max(0, \mathcal{C}_{\text{coord}}(z) - \mathcal{C}_{\text{thresh}})). \quad (42)$$

By setting $\mathcal{C}_{\text{thresh}} \approx 0.65$, the weighting function dynamically zeroes out parameter updates in vertical layers where physical profile curvature is coordinate-induced [?]. This locks the calibration loop strictly onto true constant-flux or low- ζ_{zz} regions, preventing the optimizer from corrupting local stability physics to compensate for unresolved flux-profile geometry [?].

5.3 Asymptotic Regularity Constraints on Turbulence Closures

Near the SBL height or during severe decoupling transitions, turbulent fluxes of momentum and heat vanish asymptotically as they approach the collapse height $z \rightarrow z_c^-$ [?]. We parameterize these vertical decay profiles as:

$$\tau(z) \sim \tau_0 \left(1 - \frac{z}{z_c}\right)^a \quad \text{and} \quad H(z) \sim H_0 \left(1 - \frac{z}{z_c}\right)^b. \quad (43)$$

Lemma 4 establishes that the local inverse Obukhov length behaves as $\chi(z) \sim C(z_c - z)^p$, where the regularity exponent is defined by $p = b - \frac{3}{2}a$ [?]. Standard 1.5-order closures commonly assume linear decay envelopes ($a = 1, b = 1$), yielding a negative exponent $p = -0.5$. This linear decay violates the C^2 boundary regularity threshold, introducing a formal coordinate singularity ($\chi'' \rightarrow \infty$) that manufactures unbounded diagnostic gradients and spurious coordinate-induced knees at the top of the SBL [?].

To guarantee continuous coordinate regularity and prevent these numerical artifacts, sub-grid turbulence schemes must be formulated to satisfy:

$$p = b - \frac{3}{2}a \geq 2. \quad (44)$$

By setting the decay exponents to $a = 2$ and $b = 5$, we force $p = 5 - \frac{3}{2}(2) = 2$, which satisfies the C^2 threshold and ensures that the coordinate derivatives ζ_z and ζ_{zz} remain bounded and smooth as they transition into the neutral free atmosphere [?].

6 Conclusions and Future Outlook

In this work, we have established a mathematically rigorous, singularity-free framework for stable boundary layer modeling by unifying Geometric Singular Perturbation Theory (GSPT) with the differential geometry of vertical profiles [?]. By re-parameterizing Nieuwstadt local scaling using the inverse Obukhov length profile $\chi(z) \equiv 1/L(z)$, we resolved the neutral singularity and proved that visual "knees" in physical vertical space are frequently "Fold Illusions" driven by coordinate-induced mapping curvature ($Ri_{g,zz} = C_{\text{constitutive}} + C_{\text{mapping}}$) under active vertical flux divergence, rather than genuine dynamical collapses of turbulent mixing [?, ?].

Dynamically, we mapped SBL turbulence collapse as a slow-passage bifurcation across a folded, three-

branch critical manifold, where transition to the decoupled very stable regime corresponds to a saddle-node fold where the fast eigenvalue vanishes ($\lambda_f \rightarrow 0$), destroying Fenichel normal hyperbolicity [?, ?]. To stabilize discrete NWP and SCM solvers across these delicate regions, we introduced:

1. **An Adaptive Equidistribution Grid-Stretching formulation** ($\Delta z \propto 1/|\zeta_{zz}|$): Which concentrated nodes around migrating jet noses to resolve high- ζ_{zz} gradients, achieving quadratic spatial convergence $\mathcal{O}(N_z^{-2})$ while remaining bounded within $\Delta z_{\min}$ and $\Delta z_{\max}$ to preserve CFL constraints [?, ?].
2. **Fenichel Invariant Manifold Slaving** ($u_2 = \Phi(u_1, S)$): Which replaced prognostic fast-mode ODEs with regularized algebraic slow-manifold parameterizations, eliminating spurious cubic Galerkin blowups and guaranteeing global solver stability [?, ?].
3. **Asymptotic Boundary Regularity Constraints** ($b - \frac{3}{2}a \geq 2$): Which eliminated coordinate singularities at the SBL top by forcing momentum and heat fluxes to decay quadratically and quintically, respectively, near transition boundaries [?, ?].

These theoretical and numerical formulations are directly validated against CASES-99 tower observations and GABLS3 single-column benchmarks, proving that our error-partitioning schema and gradient masking weights prevent unphysical parameter drift and stabilize solvers under non-vanishing CFL limits [?, ?].

6.1 Future Research Directions

Three promising avenues of future research emerge from this GSPT framework:

1. **Doppler Lidar Profiles:** Extending our pre-diagnostic Tikhonov splines to Doppler Lidar profiles will allow us to handle range-dependent signal-to-noise ratios (SNR), where measurement uncertainty $\delta(z)$ dynamically increases with altitude, preserving boundary-layer top gradients.
2. **WSINDy Dynamical Discovery:** Feeding our regularized, non-singular coordinate fields directly into WSINDy (Weak-form Sparse Identification of Non-linear Dynamics) will enable us

to extract candidate governing PDEs for SBL transitions from raw field data, directly evaluating our fast-slow prognostic hypotheses.

3. **ML-Based Regime Classification:** Machine learning classifiers trained on GSPT coordinate trajectories (tracking the migration of the $\zeta_z = 0$ fold locus in height-time space) can identify nocturnal transitions and decoupling events with high physical explainability.

We believe that embedding these continuous geometric constraints directly into NWP grid and solver architectures represents a critical step toward physically consistent, grid-independent stable boundary layer parameterizations.